

	more than 0.10 s.
5(a)(i)	The first minima are located at angular displacements of θ $= \sin^{-1}\left(\frac{\lambda}{b}\right)$ $= \sin^{-1}\left(\frac{590 \times 10^{-9}}{0.60 \times 10^{-3}}\right)$ $= 9.833 \times 10^{-4} \text{ rad}$ Displacement of first minima from centre of screen

	$x = 2.4 \times 9.833 \times 10^{-4}$ $= 2.36 \times 10^{-3} \text{ m}$ Width of central fringe $= 2x$ $= 2 \times 2.36 \times 10^{-3}$ $= 4.72 \times 10^{-3} \text{ m}$
5(a)(ii)	
5(b)(i)	The <i>Rayleigh criterion</i> states that for two sources to be resolved into distinct objects, the central maxima in the diffraction pattern of their images cannot be closer to each other than the distance from each to its respective first minima.
5(b)(ii)	Using $\theta = \frac{\lambda}{b}$ $= \frac{590 \times 10^{-9}}{0.60 \times 10^{-3}}$ $= 9.83 \times 10^{-4} \text{ rad}$
6(a)	Electric field strength in the parallel plate $E = \frac{V}{d}$